

CUNTU, Romania

WMO: 153160

Lat: 45.3004N

Lon: 22.5009E

Elev: 4783

StdP: 12.33

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	10.4	-5.9	4.8	15.8	0.0	6.5	18.1	35.3	24.7	27.2	21.9	6.1	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	72.2	60.1	69.8	59.0	67.6	58.2	62.8	68.9	61.2	67.4	59.7	65.7	5.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.0	95.9	64.9	59.1	89.4	63.1	57.5	84.4	62.2	31.0	69.3	29.7	67.7	28.5	65.7	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	18.6	15.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	24.8	28.0	31.8	39.7	48.4	55.7	57.9	58.6	50.5	43.4	36.4	28.4
	DBStd	14.19	8.87	8.24	8.20	9.71	6.54	6.36	5.75	6.29	8.44	8.88	9.12	8.07
	HDD50	3857	783	616	564	330	108	21	7	9	98	241	410	670
	HDD65	8407	1248	1036	1029	760	514	282	230	210	437	669	857	1135
	CDD50	954	0	0	0	20	59	193	250	277	115	37	3	0
	CDD65	29	0	0	0	0	0	3	9	13	4	0	0	0
	CDH74	41	0	0	0	0	0	0	8	27	6	0	0	0
	CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0
	WSAvg	5.2	5.2	5.9	5.0	5.3	4.9	4.6	5.2	4.0	5.0	5.0	6.0	6.3
	PrecAvg	33.6	2.0	1.9	1.9	2.5	3.4	4.1	4.4	3.0	3.5	2.6	2.1	2.2
(15) Precipitation	PrecMax	50.2	4.3	6.6	4.5	4.2	6.5	8.9	9.8	8.1	7.6	5.4	5.9	4.7
	PrecMin	23.2	0.7	0.3	0.4	0.4	1.3	1.0	1.5	0.2	0.2	0.1	0.0	0.2
	PrecStd	8.0	0.9	1.3	1.0	0.9	1.4	1.9	2.0	1.9	1.9	1.4	1.4	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.5	49.5	53.1	62.9	66.4	71.8	74.8	77.0	73.2	64.4	57.8	50.7
		MCWB	36.8	38.9	43.0	50.3	55.9	60.6	60.5	61.4	58.8	51.3	46.9	35.8
	2%	DB	41.2	44.8	49.2	59.4	63.0	69.8	71.8	72.6	68.2	60.9	52.8	44.9
		MCWB	35.2	37.7	40.7	47.8	53.9	60.6	61.0	60.1	55.5	50.1	45.2	36.3
	5%	DB	38.9	40.7	46.0	55.8	60.4	67.6	68.9	69.7	64.9	57.8	49.9	41.8
		MCWB	34.4	35.6	38.9	46.2	52.9	59.1	59.3	58.1	54.7	47.7	42.7	35.6
	10%	DB	36.4	38.1	43.0	52.3	57.8	65.0	66.4	67.1	61.6	55.1	47.4	39.0
		MCWB	33.4	34.1	37.4	44.0	51.6	57.6	58.5	57.5	53.2	46.4	42.1	34.1
	0.4%	WB	39.0	40.7	44.7	52.9	58.8	64.7	64.7	64.6	61.2	54.2	50.1	41.1
		MCDB	41.0	45.4	51.0	61.0	63.6	68.4	68.8	71.0	70.6	60.9	54.0	44.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.2	38.4	41.9	49.5	56.6	62.4	62.4	62.3	57.6	51.7	47.5	38.8
		MCDB	39.7	43.0	46.9	56.9	60.9	67.6	68.1	69.8	64.1	58.8	51.7	42.3
	5%	WB	35.4	36.9	39.9	47.3	54.4	60.5	61.0	60.3	55.8	49.9	45.1	36.6
		MCDB	37.7	41.0	45.0	54.8	58.3	65.5	67.2	67.2	62.8	56.0	48.4	39.8
	10%	WB	33.6	35.2	37.9	45.1	52.5	58.8	59.6	58.8	54.0	48.1	43.2	35.0
		MCDB	35.8	38.1	42.4	52.0	56.7	63.6	65.4	65.5	60.3	53.5	46.1	38.6
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.3	9.9	10.7	9.7	10.2	10.1	10.1	9.3	9.7	8.7	8.7
		MCDBR	10.1	10.6	12.5	13.7	12.3	12.6	12.8	12.9	12.2	12.2	10.7	10.6
		MCWBR	8.1	7.6	8.7	9.2	9.4	10.8	9.7	9.3	8.6	10.0	7.7	7.0
		MCDBR	8.8	9.9	11.4	12.9	11.1	11.6	11.3	11.9	11.4	11.5	9.4	9.4
	5% WB	MCWBR	7.8	7.9	8.7	9.1	9.9	11.1	9.7	10.0	9.1	9.8	7.6	7.9
(40) Clear-Sky Solar Irradiance	taub		0.278	0.303	0.340	0.389	0.377	0.369	0.382	0.395	0.363	0.330	0.305	0.279
	taud		2.381	2.316	2.310	2.195	2.277	2.327	2.287	2.249	2.324	2.425	2.451	2.426
	Ebn at Noon		266	278	282	275	282	285	279	271	270	265	253	253
	Edh at Noon		23	30	35	43	41	39	41	41	35	27	22	20
(44) All-Sky Solar Radiation	RadAvg		420	671	1028	1389	1657	1866	1845	1670	1188	820	483	349
	RadStd		53	87	143	156	151	178	149	165	121	98	62	47

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+1.21	N/A	N/A	N/A	N/A	N/A	-207	-317	+216	+112

Nomenclature: See separate page